

A Fast Two-Stage Algorithm for Multiple Changepoint Detection With General Segment-Wise Losses

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Abstract

We propose a general and fast two-stage framework for multiple changepoint detection based on segment-wise fitting criteria and fitted losses. The first stage constructs a loss-based gain statistic that extends the role of classical CUSUM-type scanning statistics to models with complex or high-dimensional parameters. The second stage jointly refines the initial changepoint locations while keeping the first-stage segment-wise parameter estimates fixed, thereby avoiding repeated model fitting over candidate segmentations. Simulation studies in high-dimensional mean-shift, regression, and graphical-model settings show improved localization accuracy with little additional computation, particularly when segment-wise fitting is expensive. An application to S&P 500 returns illustrates the detection of interpretable changes in a high-dimensional conditional dependence network.

Keywords: Multiple changepoint detection; loss-based gain statistic; global refinement; binary segmentation; high-dimensional models; Gaussian graphical model.

1 Introduction

Changepoint detection aims to identify time points at which the data-generating mechanism changes. Classical studies mainly focused on low-dimensional parameters, such as a univariate mean or variance. Modern applications, however, often involve much richer objects: a high-dimensional mean vector, a sparse regression coefficient, a covariance matrix, or a conditional dependence network. Examples include changes in high-dimensional mean vectors and covariance structures (Jirak, 2015; Cho, 2016; Wang and Feng, 2023),

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regression coefficients (Leonardi and Bühlmann, 2016; Lee et al., 2016; Wang et al., 2021), and Gaussian graphical models for conditional dependence networks (Gibberd and Nelson, 2017; Gibberd and Roy, 2017; Londschieen et al., 2021). These problems share a common practical difficulty. The parameter that changes is no longer a simple scalar contrast, and estimating it on each candidate segment may require regularized or otherwise computationally intensive model fitting. Consequently, a multiple changepoint procedure must address two issues at the same time: it should be flexible enough to accommodate different segment-wise models, and it should avoid excessive repeated fitting over candidate segmentations.

A standard way to formulate multiple changepoint detection is through a segment-wise loss. For any candidate interval, one fits the model on that interval and evaluates the resulting fitted loss. Dynamic-programming-based procedures can then search over candidate partitions and provide a globally optimized segmentation (Leonardi and Bühlmann, 2016). This strategy is conceptually natural, but it can be expensive when the number of candidate intervals is large or when each segment fit involves high-dimensional regularization. Binary segmentation and its variants, including wild binary segmentation and seeded binary segmentation, are more attractive computationally because they reduce a multiple changepoint problem to a sequence of single-changepoint searches (Scott and Knott, 1974; Venkatraman, 1992; Fryzlewicz, 2014; Kovács et al., 2023). Their use, however, depends on a scanning statistic that measures the evidence for a change at each candidate split. In the classical mean-shift model, this statistic is usually a CUSUM contrast (Page, 1954; Hinkley, 1970; Csörgő and Horváth, 1997). For regression models, graphical models, or other structured models, there is no equally automatic CUSUM-type statistic because the object being compared is itself estimated from the data.

This observation motivates the first component of our framework. Instead of designing a model-specific contrast for each setting, we measure the evidence for a split by the reduction in fitted loss. Specifically, for an interval $(s, e]$ and a candidate split t , the proposed gain statistic compares the fitted loss on the full interval with the sum of the fitted losses on the two subintervals. Thus, a large gain means that two segment-wise fits explain the data substantially better than one common fit. In a univariate mean-shift model, this loss reduction reduces to the squared CUSUM statistic up to scaling. Therefore, the proposed statistic retains the intuition of CUSUM while extending it to settings where the changing parameter may be a vector, a regression coefficient, or a network structure. Once a segment-wise fitting criterion and a fitted loss are specified, the same gain statistic can be used with binary-segmentation-based procedures.

A second issue arises after initial detection. Fast procedures such as binary segmentation are useful for producing preliminary estimates, but their estimated changepoint locations may still be inaccurate, especially when the model is high-dimensional or the signal is

distributed across many coordinates or edges. A common remedy is local refinement, which updates each estimated changepoint within a neighborhood determined by the initial segmentation; see, for example, [Xu et al. \(2023\)](#) and [Kaul et al. \(2023\)](#). Such refinements are computationally convenient, but they remain local and depend on the initial neighborhoods. At the other extreme, one could refine all boundaries by minimizing a fully profiled global loss. This would jointly optimize the changepoint locations, but it would also require refitting the segment model repeatedly over many candidate segments, which is precisely the computational bottleneck in high-dimensional regression and graphical-model applications.

We therefore propose a fast two-stage framework. In the first stage, the loss-based gain statistic is combined with a segmentation procedure, such as wild binary segmentation, to estimate the number and locations of changepoints and to obtain the corresponding segment-wise parameter estimates. In the second stage, we keep these first-stage parameter estimates fixed and jointly re-optimize all estimated changepoint locations. This fixed-parameter global refinement separates the expensive task of parameter estimation from the cheaper task of boundary optimization. It is global because all estimated changepoints are optimized jointly under the ordering constraint, but it is fast because evaluating the refinement objective only requires fitted-loss calculations at fixed parameter values. The refinement step can therefore be viewed as a computationally lightweight approximation to fully refitted global optimization.

The contribution of this paper is threefold. First, we introduce a loss-based gain statistic that provides a unified scanning rule for multiple changepoint detection with general segment-wise losses. This makes it possible to use the same detection principle across mean-shift, regression, graphical-model, and other structured settings, without deriving a separate CUSUM-type statistic for each model. Second, we propose a fixed-parameter global refinement step that improves initial localization by jointly optimizing all detected boundaries while avoiding repeated high-dimensional model fitting. Third, we provide an empirical evaluation of the framework in high-dimensional mean-shift, high-dimensional regression, and Gaussian graphical-model settings. The simulation results show that the refinement step consistently reduces localization error, while its additional computational cost is small relative to the initial detection cost, especially when segment-wise fitting is expensive. An application to S&P 500 return data further illustrates how the method can detect interpretable changes in high-dimensional conditional dependence networks.

The remainder of the paper is organized as follows. [Section 2](#) introduces the general loss-based gain statistic and the proposed two-stage procedure. [Section 3](#) reports the simulation results and computational comparisons. [Section 4](#) presents the real data example. [Section 5](#) concludes the paper.

2 A fast two-stage framework for changepoint detection

2.1 Model setup

Let $\{\mathcal{O}_t\}_{t=1}^n$ denote a sequence of observations, where $\mathcal{O}_t$ may be a scalar observation, a high-dimensional vector, or a covariate–response pair. We consider a multiple changepoint setting with unknown locations

$$0 = \tau_0 < \tau_1 < \cdots < \tau_K < \tau_{K+1} = n,$$

such that the parameter of interest is constant within each segment but may change across adjacent segments. Specifically, for $I_j = (\tau_{j-1}, \tau_j] = \{\tau_{j-1} + 1, \dots, \tau_j\}$, $j = 1, \dots, K + 1$, the observations in I_j are characterized by a segment-specific parameter $\theta_j \in \Theta$. The goal is to estimate K and the changepoint set $\mathcal{T} = \{\tau_1, \dots, \tau_K\}$.

Our framework applies whenever the parameter of interest can be estimated on a candidate interval through a segment-wise fitting criterion. For any interval $(s, e]$, let

$$\hat{\theta}_{s:e} \in \arg \min_{\theta \in \Theta} Q_n((s, e], \theta), \quad 0 \leq s < e \leq n,$$

where $Q_n((s, e], \theta)$ may be an empirical loss or a regularized empirical loss in high-dimensional settings. Let $L_n((s, e], \theta)$ denote the fitted loss used to compare candidate segmentations, and define $Z_n(s, e) = L_n((s, e], \hat{\theta}_{s:e})$. In unpenalized models, one may take $Q_n = L_n$, so that $Z_n(s, e) = \inf_{\theta \in \Theta} L_n((s, e], \theta)$. In regularized models, Q_n may include a penalty used for estimation, while L_n evaluates the corresponding fitted loss without the penalty term.

The formulation includes the following representative examples.

Mean-shift model. Let $\mathcal{O}_t = Y_t \in \mathbb{R}^p$ and $Y_t = \mu_j + \varepsilon_t$ for $t \in I_j$, where $\mu_j \in \mathbb{R}^p$ is the segment mean. Thus, $\theta_j = \mu_j$, and one may take

$$Q_n((s, e], \mu) = L_n((s, e], \mu) = \frac{1}{n} \sum_{t=s+1}^e \|Y_t - \mu\|_2^2.$$

The univariate case $p = 1$ recovers the classical mean-shift setting, in which the associated gain statistic is equivalent to a scaled squared CUSUM contrast.

High-dimensional regression model. Let $\mathcal{O}_t = (X_t, Y_t)$, where $X_t \in \mathbb{R}^p$ and $Y_t = X_t^\top \beta_j + \varepsilon_t$ for $t \in I_j$. The parameter of interest is $\theta_j = \beta_j \in \mathbb{R}^p$, where p may be large. We

use the regularized fitting criterion and fitted loss

$$Q_n((s, e], \beta) = \frac{1}{n} \sum_{t=s+1}^e (Y_t - X_t^\top \beta)^2 + \lambda_{s:e} \|\beta\|_1, \quad L_n((s, e], \beta) = \frac{1}{n} \sum_{t=s+1}^e (Y_t - X_t^\top \beta)^2.$$

High-dimensional Gaussian graphical model. Let $\mathcal{O}_t = Y_t \in \mathbb{R}^p$, with $Y_t \sim N_p(0, \Sigma_j)$ for $t \in I_j$ and $\Omega_j = \Sigma_j^{-1}$. The precision matrix Ω_j represents the conditional dependence structure. We estimate this structure through nodewise regressions, as in neighborhood selection (Meinshausen and Bühlmann, 2006). Let $B_j = (\beta_{j,1}, \dots, \beta_{j,p})$, where $\beta_{j,k} \in \mathbb{R}^{p-1}$ is the coefficient vector for predicting the k th component from the remaining components. Thus, $\theta_j = B_j$. The nodewise-Lasso fitting criterion and corresponding fitted loss are

$$Q_n((s, e], B) = \frac{1}{n} \sum_{t=s+1}^e \sum_{k=1}^p (Y_{t,k} - Y_{t,-k}^\top \beta_k)^2 + \lambda_{s:e} \sum_{k=1}^p \|\beta_k\|_1,$$

$$L_n((s, e], B) = \frac{1}{n} \sum_{t=s+1}^e \sum_{k=1}^p (Y_{t,k} - Y_{t,-k}^\top \beta_k)^2,$$

where $Y_{t,k}$ is the k th component of Y_t and $Y_{t,-k}$ is obtained by removing that component.

These examples show that the changing parameter may be a high-dimensional mean vector, a regression coefficient vector, or a conditional dependence structure. The proposed framework requires only a segment-wise fitting criterion and a fitted loss for comparing candidate segmentations, rather than a separately constructed CUSUM statistic for each model.

2.2 Gain-based formulation

A general approach to multiple changepoint detection is to optimize a segmentation objective over candidate partitions. For a candidate number of changepoints k and an ordered segmentation $0 = \tau_0 < \tau_1 < \dots < \tau_k < \tau_{k+1} = n$, define

$$J_{n,k}(\tau_1, \dots, \tau_k) = \sum_{j=0}^k Z_n(\tau_j, \tau_{j+1}).$$

When $Q_n = L_n$, this objective reduces to the fully profiled loss $J_{n,k}(\tau_1, \dots, \tau_k) = \sum_{j=0}^k \inf_{\theta \in \Theta} L_n((\tau_j, \tau_{j+1}], \theta)$.

For each k , the minimizer of $J_{n,k}$ over all ordered segmentations can be computed by dynamic programming (Leonardi and Bühlmann (2016)). The number of changepoints may then be selected by minimizing a penalized objective over $k = 0, 1, \dots, K_{\max}$, for example

$$\hat{K} = \arg \min_{0 \leq k \leq K_{\max}} \left\{ \min_{0 < \tau_1 < \dots < \tau_k < n} J_{n,k}(\tau_1, \dots, \tau_k) + \text{pen}_n(k) \right\}.$$

Although this approach yields a globally optimized segmentation, it requires segment-wise fitting over many candidate intervals and can therefore be expensive in high-dimensional or structured models.

A computationally attractive alternative is provided by binary segmentation and its variants, including wild binary segmentation (Scott and Knott, 1974; Venkatraman, 1992; Fryzlewicz, 2014). These procedures recursively search for changepoints on candidate intervals and determine their number through a thresholding rule. Their implementation, however, requires a scanning statistic that quantifies the evidence for a change at each candidate split. In classical mean-shift models, such a statistic is readily constructed through a CUSUM contrast. For regression, graphical, and other structured models, an analogous model-specific contrast is less immediate because the changing object is itself estimated from the data.

We therefore define a common loss-based scanning statistic. For an interval $(s, e]$ and a candidate split point t with $s < t < e$, let

$$\mathcal{G}_n(t; s, e) = Z_n(s, e) - Z_n(s, t) - Z_n(t, e).$$

The statistic measures the improvement in fitted loss obtained by splitting $(s, e]$ at t . A large value of $\mathcal{G}_n(t; s, e)$ indicates that two segment-wise fits provide a substantially better representation of the observations than a single fit over the full interval.

The gain statistic extends the classical CUSUM principle. To see this, consider the univariate special case of the mean-shift model, corresponding to $p = 1$ in the formulation above, with $L_n((s, e], \mu) = n^{-1} \sum_{r=s+1}^e (Y_r - \mu)^2$. Writing $\bar{Y}_{a:b} = (b - a + 1)^{-1} \sum_{r=a}^b Y_r$, we obtain

$$\mathcal{G}_n(t; s, e) = \frac{1}{n} \frac{(t - s)(e - t)}{e - s} (\bar{Y}_{s+1:t} - \bar{Y}_{t+1:e})^2,$$

which is equivalent to the squared CUSUM contrast up to scaling. Thus, maximizing $\mathcal{G}_n(t; s, e)$ recovers the classical single-changepoint scan in this setting.

The same construction applies when the segment-wise parameter is more complex. In a regression model, $\mathcal{G}_n(t; s, e)$ measures the **improvement** from allowing different coefficient vectors on the two subintervals, while in a graphical model it measures the improvement from allowing different conditional dependence structures. Hence, once Q_n and L_n are specified, the gain statistic can be used with binary-segmentation-based procedures to obtain initial multiple changepoint estimates without constructing a separate scanning statistic for each model.

For unpenalized models with $Q_n = L_n$, fitting separate models on two subintervals cannot increase the profiled loss, and hence $\mathcal{G}_n(t; s, e) \geq 0$. For regularized models, $\mathcal{G}_n(t; s, e)$ compares fitted losses evaluated at the corresponding regularized segment estimates. This formulation accommodates both classical unpenalized models and high-dimensional regu-

larized estimation within the same detection framework.

2.3 Initial detection

We apply a binary-segmentation-based procedure, such as binary segmentation or wild binary segmentation, using the gain statistic $\mathcal{G}_n$ and a threshold ζ_n . This yields an initial changepoint set $\widehat{\mathcal{T}}^{(0)} = \{\widehat{\tau}_1^{(0)}, \dots, \widehat{\tau}_{\widehat{K}}^{(0)}\}$, where $\widehat{K}$ is the estimated number of changepoints. Setting $\widehat{\tau}_0^{(0)} = 0$ and $\widehat{\tau}_{\widehat{K}+1}^{(0)} = n$, we compute the corresponding segment-wise parameter estimates as

$$\widehat{\theta}_j^{(0)} \in \arg \min_{\theta \in \Theta} Q_n((\widehat{\tau}_{j-1}^{(0)}, \widehat{\tau}_j^{(0)}], \theta), \quad j = 1, \dots, \widehat{K} + 1.$$

These first-stage estimates are used in the global refinement step.

2.4 Global refinement without repeated model fitting

Given the initial changepoint estimates $\widehat{\mathcal{T}}^{(0)}$ and the corresponding segment-wise parameter estimates $\widehat{\theta}_1^{(0)}, \dots, \widehat{\theta}_{\widehat{K}+1}^{(0)}$, we refine the changepoint locations while keeping the estimated number $\widehat{K}$ fixed. A fully refitted global refinement would minimize $J_{n, \widehat{K}}$ over all ordered segmentations with $\widehat{K}$ changepoints. Although this approach jointly optimizes the segment boundaries, it requires fitting the model on each candidate segment and can therefore be computationally expensive in high-dimensional or structured models.

We instead reuse the first-stage parameter estimates and optimize only over the changepoint locations. For an ordered segmentation $0 = \tau_0 < \tau_1 < \dots < \tau_{\widehat{K}} < \tau_{\widehat{K}+1} = n$, define

$$J_n^{\text{fix}}(\tau_1, \dots, \tau_{\widehat{K}}) = \sum_{j=0}^{\widehat{K}} L_n((\tau_j, \tau_{j+1}], \widehat{\theta}_{j+1}^{(0)}).$$

The refined estimator is

$$(\widehat{\tau}_1^r, \dots, \widehat{\tau}_{\widehat{K}}^r) \in \arg \min_{0 < \tau_1 < \dots < \tau_{\widehat{K}} < n} J_n^{\text{fix}}(\tau_1, \dots, \tau_{\widehat{K}}), \quad \widehat{\mathcal{T}}^r = \{\widehat{\tau}_1^r, \dots, \widehat{\tau}_{\widehat{K}}^r\}.$$

The refinement is global because all estimated boundaries are optimized jointly under the ordering constraint. It is computationally efficient because evaluating J_n^{fix} requires only fitted-loss evaluations at fixed parameter values, rather than repeated segment-wise model fitting. When the first-stage segmentation and parameter estimates are informative, J_n^{fix} provides a computationally inexpensive approximation to the refitted global objective.

2.5 Algorithm

The proposed two-stage procedure is summarized in Algorithm 1. For a given minimum segment length Δ_n , define the feasible set for refinement as

$$\mathfrak{S}_{\hat{K}}(\Delta_n) = \left\{ (\tau_1, \dots, \tau_{\hat{K}}) : 0 = \tau_0 < \tau_1 < \dots < \tau_{\hat{K}} < \tau_{\hat{K}+1} = n, \tau_{j+1} - \tau_j \geq \Delta_n, j = 0, \dots, \hat{K} \right\}.$$

Algorithm 1 General gain-based detection with global refinement

Require: Observations $\{\mathcal{O}_t\}_{t=1}^n$; fitting criterion Q_n ; fitted loss L_n ; initial detection method $\mathcal{A}$ with threshold ζ_n and other tuning parameters; minimum refinement segment length Δ_n

Ensure: Refined changepoint set $\hat{\mathcal{T}}^r$

1: Apply $\mathcal{A}$ with gain statistic $\mathcal{G}_n$ to obtain

$$\hat{\mathcal{T}}^{(0)} = \{\hat{\tau}_1^{(0)}, \dots, \hat{\tau}_{\hat{K}}^{(0)}\}.$$

2: Set $\hat{\tau}_0^{(0)} = 0$ and $\hat{\tau}_{\hat{K}+1}^{(0)} = n$, and compute

$$\hat{\theta}_j^{(0)} \in \arg \min_{\theta \in \Theta} Q_n((\hat{\tau}_{j-1}^{(0)}, \hat{\tau}_j^{(0)}], \theta), \quad j = 1, \dots, \hat{K} + 1.$$

3: Define

$$J_n^{\text{fix}}(\tau_1, \dots, \tau_{\hat{K}}) = \sum_{j=0}^{\hat{K}} L_n((\tau_j, \tau_{j+1}], \hat{\theta}_{j+1}^{(0)}).$$

4: Compute

$$(\hat{\tau}_1^r, \dots, \hat{\tau}_{\hat{K}}^r) \in \arg \min_{(\tau_1, \dots, \tau_{\hat{K}}) \in \mathfrak{S}_{\hat{K}}(\Delta_n)} J_n^{\text{fix}}(\tau_1, \dots, \tau_{\hat{K}}).$$

5: **return** $\hat{\mathcal{T}}^r = \{\hat{\tau}_1^r, \dots, \hat{\tau}_{\hat{K}}^r\}.$

2.6 Implementation and computation of global refinement

Because the first-stage parameter estimates are fixed, the refinement objective can be minimized globally by dynamic programming without refitting the segment model. For $k = 0, \dots, \hat{K}$ and $v \leq n$, define

$$F_k(v) = \min_{\substack{0=\tau_0 < \tau_1 < \dots < \tau_k < \tau_{k+1}=v \\ \tau_{j+1}-\tau_j \geq \Delta_n, j=0, \dots, k}} \sum_{j=0}^k L_n((\tau_j, \tau_{j+1}], \hat{\theta}_{j+1}^{(0)}).$$

The initialization is $F_0(v) = L_n((0, v], \hat{\theta}_1^{(0)})$ for $v \geq \Delta_n$, and the recursion is

$$F_k(v) = \min_{k\Delta_n \leq u \leq v - \Delta_n} \left\{ F_{k-1}(u) + L_n((u, v], \hat{\theta}_{k+1}^{(0)}) \right\}, \quad k = 1, \dots, \hat{K}.$$

The refined changepoints are recovered by backtracking from $F_{\hat{K}}(n)$. When L_n is additive over observations, the loss terms in the recursion can be computed from cumulative losses evaluated at the fixed estimates $\hat{\theta}_1^{(0)}, \dots, \hat{\theta}_{\hat{K}+1}^{(0)}$.

To compare computational costs, let $\mathbf{C}_{\text{fit}}(m, p)$ denote the cost of fitting the segment-specific model on a segment of length m and dimension p . For standard binary segmentation, a balanced recursion entails $O\{n \log(n)\}$ candidate split evaluations, yielding the schematic first-stage cost

$$O\{n \log(n) \mathbf{C}_{\text{fit}}(n, p)\}.$$

Variants such as wild binary segmentation have a cost determined by their candidate interval collection, but likewise require repeated segment-wise fitting.

In contrast, once the first-stage parameter estimates have been obtained, the dynamic programming refinement requires $O(\hat{K}n^2)$ fixed-loss evaluations, after cumulative losses have been constructed. Thus, for fixed or moderate $\hat{K}$, its location-optimization cost is $O(n^2)$. Most importantly, **the refinement step avoids the repeated model-fitting cost $\mathbf{C}_{\text{fit}}(n, p)$** . This distinction is particularly consequential in high-dimensional regression and graphical-model settings, where segment-wise fitting is substantially more expensive than evaluating a fitted loss at fixed parameter values.

3 Simulation studies

We evaluate the localization accuracy and computational cost of the proposed procedure in three model settings.

3.1 Simulation settings

We consider a high-dimensional mean-shift model, a high-dimensional regression model, and a high-dimensional Gaussian graphical model. Throughout, $n = 600$ and the true changepoint set is $\mathcal{T} = \{150, 300, 450\}$. Initial estimates are obtained by wild binary segmentation using the corresponding gain statistic, followed by the proposed global refinement. Localization results are based on 100 replications, and computing times are averaged over five replications.

High-dimensional mean-shift model. We generate $Y_t = \mu_j + \varepsilon_t$ for $t \in I_j$, where

Table 1: Localization accuracy over 100 replications. Entries are mean scaled Hausdorff distances, with standard deviations in parentheses.

Model	Initial d_H/n	Refined d_H/n	Improvement	$\mathbb{P}(\hat{K} = K)$
Mean-shift model	0.0024 (0.0033)	0.0021 (0.0023)	9.8%	1.00
Regression model	0.0053 (0.0248)	0.0045 (0.0249)	15.7%	0.99
Graphical model	0.0344 (0.0840)	0.0326 (0.0845)	5.0%	0.87

$Y_t \in \mathbb{R}^{50}$ and $\varepsilon_t \sim N_{50}(0, I_{50})$. Only the first 10 coordinates change across segments, with

$$(\mu_{1k}, \mu_{2k}, \mu_{3k}, \mu_{4k}) = (0, 1.5, -1, 1), \quad k = 1, \dots, 10,$$

and $\mu_{jk} = 0$ for $k = 11, \dots, 50$ and all j .

High-dimensional regression model. We generate $Y_t = X_t^\top \beta_j + \varepsilon_t$ for $t \in I_j$, where $X_t \sim N_{50}(0, I_{50})$ and $\varepsilon_t \sim N(0, 1)$. The coefficient vectors are sparse, with nonzero entries of magnitude 1.5 and segment-specific active sets and signs.

High-dimensional graphical model. We generate $Y_t \sim N_{20}(0, \Sigma_j)$ for $t \in I_j$, where $\Omega_j = \Sigma_j^{-1}$ is sparse. The four segments correspond to chain, banded, hub, and block-with-bridge graph structures, respectively.

Evaluation metrics and implementation details. Localization accuracy is measured by the scaled Hausdorff distance $d_H(\hat{\mathcal{T}}, \mathcal{T})/n$, with smaller values indicating more accurate localization. We also report $\mathbb{P}(\hat{K} = K)$. For initial detection, we use wild binary segmentation with $M \in [30, 80]$ random intervals and a minimum segment length of 30; the same minimum segment length is used in the global refinement step. For the mean-shift and graphical-model settings, the detection threshold is of the form $\zeta_n = c_\zeta \frac{p \log(p)}{n}$, where $c_\zeta \in (0.5, 1.5)$ for the mean-shift model and $c_\zeta \in (0, 1)$ for the graphical model. For the regression setting, we use $\zeta_n = c_\zeta \frac{\log(p)}{n}$, $c_\zeta \in (3, 5)$. For the regularized regression and graphical-model fits, the penalty level takes the form $\lambda_{s:e} = c_\lambda \sqrt{\frac{\log(p)}{e-s+1}}$, with $c_\lambda \in (2, 4)$. Within each model setting, the selected tuning parameters are held fixed across replications.

3.2 Generality and effect of refinement

Figure 1 displays the scaled Hausdorff distances before and after global refinement over 100 replications, and Table 1 reports the corresponding numerical summaries together with the probability of selecting the correct number of changepoints.

The refinement step reduces the mean localization error in all three settings. The largest relative improvement is observed in the high-dimensional regression model, whereas the mean-shift model is already accurate before refinement. The graphical-model setting has the largest mean localization error, reflecting the greater difficulty of detecting changes

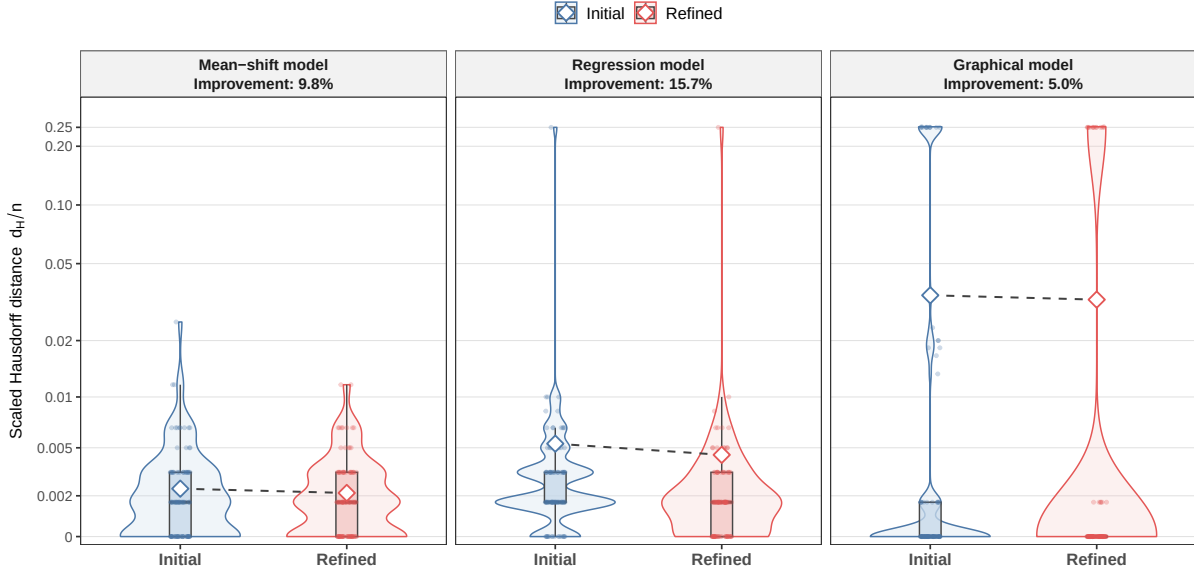


Figure 1: Scaled Hausdorff distances before and after global refinement over 100 replications. The violin plots and embedded boxplots display the replication-level distributions, while diamonds indicate the mean values reported in Table 1. The improvement in each panel is the percentage reduction in the mean localization error after refinement. The vertical axis is displayed on a pseudo-logarithmic scale.

Table 2: Average computing times(seconds) over five replications. The last column reports the global refinement time as a percentage of the initial detection time.

Model	Initial detection	Global refinement	Ref./Init.
mean-shift model	12	0.80	6.67%
Regression model	180	0.83	0.46%
Graphical model	978	0.90	0.09%

in conditional dependence structures. Its distribution also exhibits a small number of relatively large errors, which increase the overall mean despite many replications having small localization errors.

3.3 Computational cost

Finally, we evaluate the additional computational cost of the refinement step. Table 2 reports the average computing time for the initial detection step and the refinement step. We also report the ratio between the refinement time and the initial detection time.

Figure 2 visualizes the computing times of the two stages. Because the computing times differ substantially across model settings, the vertical axis is displayed on a pseudo-logarithmic scale; the corresponding times on the original scale are reported in Table 2. For

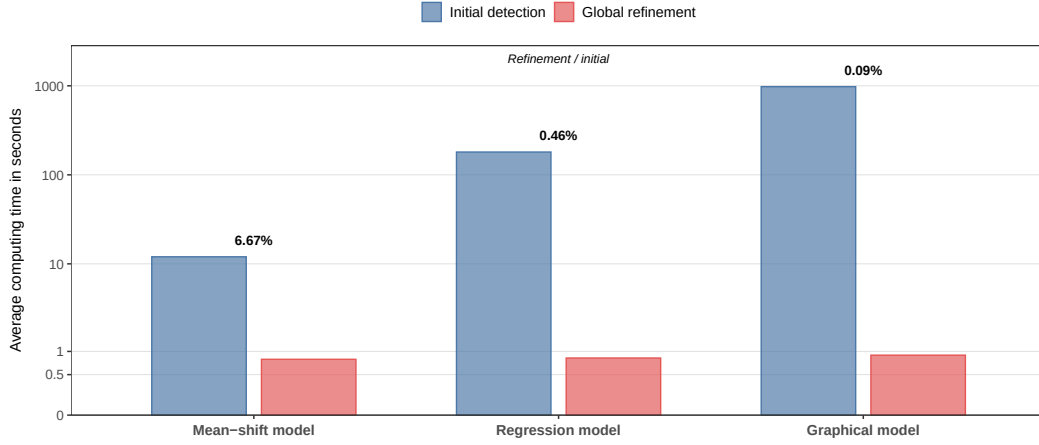


Figure 2: Average computing times of the initial detection and global refinement steps over five replications. The vertical axis is displayed on a pseudo-logarithmic scale to show the large differences in computing time across models. The annotation above each model gives the refinement time as a percentage of the initial detection time.

Table 3: Initial and refined changepoint estimates for the selected S&P 500 stock returns.

Period	Initial date	Refined date	Market episode
2007–2011	2008-10-16	2008-10-14	Global financial crisis
	2010-04-26	2011-11-14	European sovereign debt crisis
2019–2024	2020-03-25	2020-03-26	COVID-19 market shock

the mean-shift model, the high-dimensional regression and graphical-model settings, the global refinement step requires only 6.67%, 0.46% and 0.09% of the initial detection time, respectively. This supports the main computational advantage of the proposed method: global refinement avoids repeated fitting of expensive high-dimensional segment models.

4 Real data example

We apply the proposed procedure to daily returns of 50 selected S&P 500 stocks from six sectors: technology, financials, healthcare, consumer, industrials, and energy/utilities. Two periods are analyzed separately: 2007–2011, covering the global financial crisis and the European sovereign debt crisis, and 2019–2024, covering the COVID-19 market shock and the subsequent period. Within each period, the returns are standardized columnwise, and the conditional dependence structure is modeled by a Gaussian graphical model fitted using nodewise Lasso. Table 3 reports the initial and refined changepoint estimates obtained by the proposed procedure.

The refined changepoints correspond to three major episodes of market instability:

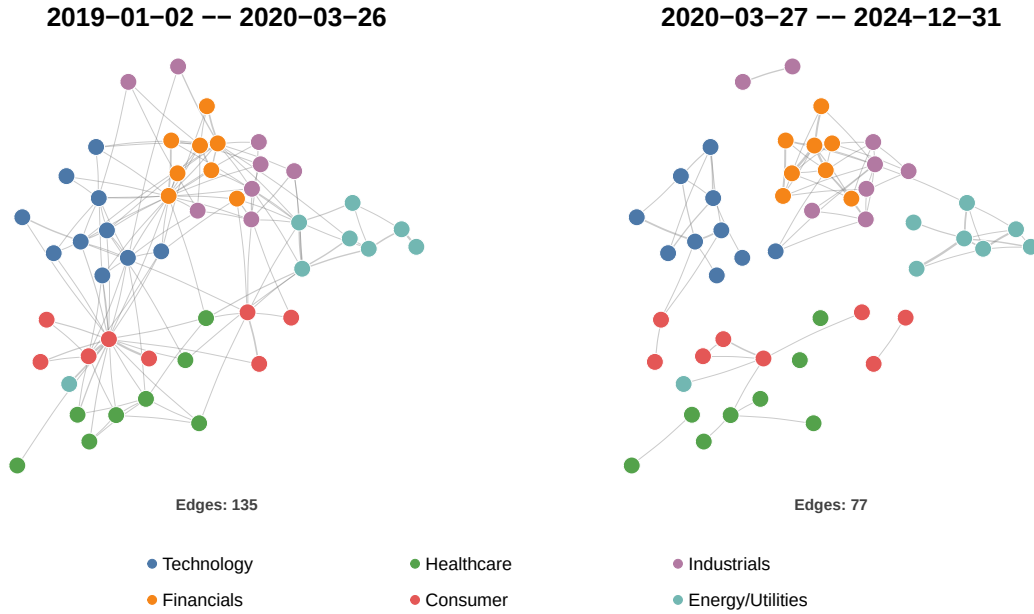


Figure 3: Estimated conditional dependence networks of the selected S&P 500 stocks before and after the refined changepoint in the 2019–2024 period. Nodes are colored by sector, and edges represent estimated conditional dependence relationships.

the global financial crisis, the European sovereign debt crisis, and the COVID-19 market shock. For the 2019–2024 period, Figure 3 compares the estimated conditional dependence networks before and after the refined changepoint in March 2020. The estimated network after the changepoint is substantially sparser, indicating a marked change in conditional dependence among the selected stocks.

5 Conclusion

We proposed a general and fast two-stage framework for multiple changepoint detection based on segment-wise fitting criteria and fitted losses. The proposed gain statistic provides a common scanning rule for obtaining initial changepoint estimates across different model classes, while the fixed-parameter global refinement jointly improves their locations without repeated segment-wise model fitting. Simulation studies illustrate improved localization in high-dimensional mean-shift, high-dimensional regression, and graphical-model settings, with particularly small additional computational cost when model fitting is expensive. The S&P 500 application further demonstrates the detection of interpretable changes in a high-dimensional conditional dependence network. As the refinement step keeps the estimated number of changepoints fixed, its performance depends on the quality of the initial detection result.

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